1st Semester MFE209TC: Sensors, Instrumentation and Industrial Applications
School of Intelligent Manufacturing Ecosystem, Dr. Yasunori Sakai

Team Project

Design, Prototyping, and Evaluation of Flexible Sensors using DIW 3D Printing (Development of Improved Sensors Based on Literature Surveys)

1. Objective

In this project, students will use a DIW (Direct Ink Writing) 3D printer to prototype two types of flexible sensors: capacitive and piezoresistive. Students will then evaluate their performance.

Teams will independently execute the following process:

- Survey existing academic literature to identify areas for improvement.
- Design the sensor structure based on improvement ideas.
- Prototype using DIW.
- Evaluate using a simple measurement system based on Arduino.
- Refine the design based on evaluation results and re-evaluate.
- Present and report the final sensor design as students team's project outcome paper.

2. Learning Outcomes (LOs)

This assignment will assess ALL learning outcomes (A-E) of the module.

LO	Description
A	Demonstrate an understanding of the physical principles of sensors and apply them to practical designs.
B	Analyze and interpret sensor specifications (sensitivity, resolution, linearity, responsiveness).
C	Examine and evaluate requirements for measurement systems (sensor + circuit + data acquisition).
D	Evaluate and analyze factors affecting sensor performance (materials, shape, noise).
E	Design, prototype, and evaluate actual sensors, proposing them for specific applications.

3. Available Materials and Equipment

Item	Details
3D Printer	DIW type (SenGong AutoBio2000), A1003 Dynamics Lab2 With Slicer software CURA.
Substrate	Polyimide film (thickness 50–125 μm)
Conductive Material	Conductive paste for DIW
Dielectric (for Capacitive)	Soft silicone rubber
Measurement Instruments	Arduino board, breadboard, resistors, capacitor, coil and jumper wires, etc.
Evaluation Tools	Digital multimeter, force sensor, weight set, etc.

4. Application Scenarios (Select one per team)

No.	Scenario	Target Measurement	Role of Capacitive Sensor	Role of Resistive Sensor
1	Push Force Sensor	Pressure (Vertical Load)	Capacitance change due to pressure	Resistance change due to compression
2	Bending Sensor	Bending Angle / Displacement	Distance change between electrodes	Resistance change due to stretching
3	Touch & Release Sensor	Detection of Contact /	Capacitance change upon	Vibration detection upon release

No.	Scenario	Target Measurement	Role of Capacitive Sensor	Role of Resistive Sensor
		Release	proximity/contact	
4	Vibration Sensor	Vibration / Impact	Electrode distance change during vibration	Micro-stretch resistance change during vibration

5. Assignment Details (Mandatory Items)

1) Literature Survey & Improvement Concept (LO A, E)

- Survey 10–20 academic papers on existing flexible sensors as a team.
- Identify *current limitations* and *room for improvement* from the literature.
- Clarify student's sensor design concept based on these improvement ideas.
- Explain why this improvement is effective based on physical principles.

2) Sensor Design (Structure & Materials) (LO A, C)

- Design a specific sensor structure to realize the improvement concept.
- Electrode shape (serpentine pattern, circular patch, parallel plates, etc.).
- Layer configuration (stacking of electrode / dielectric / substrate).
- Design parameters (line width, gap, thickness, area, etc.).
- Determine the materials and their formulations to be used.
- Create design drawings using CAD software (Solidworks, Rhino, Autodesk Fusion360, etc. All resources are available in XJTLU).

3) Prototyping Process (LO E)

- Document the prototyping process using the DIW 3D printer.
- Printing conditions (nozzle diameter, extrusion pressure, print speed, curing temperature/time).
- Method of connecting electrodes (attaching lead wires).
- Record any problems encountered during prototyping and how students addressed them.
- Report both successes and failures honestly (failures do not incur penalties).

4) Evaluation Experiments & Data Analysis (LO B, D)

- Measure the sensor response using an Arduino-based system.

- Capacitive: RC time constant method (Arduino CapacitiveSensor library recommended).
- Resistive: Voltage divider circuit (Arduino analog input).
- Clearly describe the evaluation protocol details (procedure).
- What forces were applied? (Weights, finger pressing, bending, tapping).
- How was data recorded? (Sampling rate, number of measurements).
- Evaluate the following sensor characteristics: Sensitivity, Linearity, Responsiveness, Repeatability.
- How to calibrate a self-made sensor output (discuss and provide feasible method)?

5) Refinement Cycle (LO D, E)

- Modify the design based on the results of the initial evaluation.
- Adjustments to structure, shape, dimensions, or material formulation.
- Conduct re-prototyping and re-evaluation.
- Clearly show comparisons before and after the improvements using graphs or tables.
- Discuss how the improvements affected performance.

6) Comprehensive Discussion & Application Proposal (LO C, E)

- Provide a comprehensive performance evaluation of the developed sensor.
- Propose specific application methods based on the chosen scenario.
- Discuss current limitations and potential for future improvements.
- Compare proposed sensor with existing sensors (in terms of cost, performance, application and manufacturability).

6. Basic Circuits for Measurement (Detailed info will be provided in the lecture)

Measuring Capacitive Sensors (RC Time Constant Method)

- Insert a resistor between Arduino digital pins.
- Measure the charging/discharging time of the sensor (capacitor).
- The CapacitiveSensor library for Arduino is recommended for detecting the signals.

Measuring Resistive Sensors (Voltage Divider)

- Connect the sensor (variable resistor) and a resistor in series to an Arduino analog input pin.
- Measure the divided voltage value.
- Detect resistance changes as voltage changes.

7. Submission Requirements

Item	Details
Report Deadline	October 23, 2026 (Friday) 17:00
Pitch Presentation	Final Week (Week 6: Sometime between Oct 26–30 during class time). Details TBA based on the class size
Report Length	10–30 pages (including figures, tables, not including and references)
Presentation Time	5-10 min presentation + several mins Q&A (All team members must present), Details TBA based on the class size If a student is absent without a valid reason officially approved by the University, a penalty of 50 points will be deducted from the student's team assignment mark.
Format	For report IEEE International Conference Template (MS Word / LaTeX) Template file have been uploaded on LM core's module page. Detailed formatting directions are written in the template and students need to follow all things for writing professional report. For presentation slide: No requirement for formatting. Each team can make it more creative and informative.
File Format	PDF and pptx
File Name (Report)	TeamName_Project_Report.pdf
File Name (Slide)	TeamName_Project_Slide.pptx
Method	Upload to the relevant page on Learning Mall Core

Item	Details
References	IEEE style. Datasheets and online materials may be cited. - Do NOT allow to use WEB pages for the reference. - All reference need to have DOI information - All reference need to be referred in the main text of your report with appropriate referencing method in IEEE style

Important note: Late submissions without university-allowed valid reasons will not be accepted. Penalties will be applied in accordance with the University's Code of Practice on Assessment

8. Rules for AI Usage

The same rules apply as for the individual assignment.

- An AI Usage Declaration must be included at the beginning of the report.
- AI-generated content is limited to 10% of the overall text.
- When using AI-generated text directly, it must be explicitly marked with quotation marks (" "), and the AI tool name and usage details must be listed in the declaration.
- Experimental data, prototyping records, discussions, and conclusions must be student's own original writing.
- False declarations will result in strict academic misconduct penalties.

9. Assessment Criteria (Rubric)

Criteria	LO	Weight	Description
Sensor Principles & Design Concept	A	20%	Correct understanding of physics, validity of the improvement concept. Does the survey inform the design?
Specification Analysis & Performance Evaluation	B, D	20%	Appropriateness of evaluation protocol, data analysis quality, and discussion of performance factors (noise, environment).

Criteria	LO	Weight	Description
System Design (Structure, Process, Refinement)	C	15%	Rationality of the sensor structure and prototyping process. Is the PDCA refinement cycle clear?
Practical Application & Proposal	E	15%	Were prototyping and evaluation successfully completed? Are the application proposals realistic and persuasive?
Report Quality (Structure, Figures, IEEE format)	ALL	10%	Overall structure, clarity of figures, and adherence to IEEE formatting.
Final Week Presentation (Content & Delivery)	ALL	10%	Clarity, persuasiveness, time management, and handling of Q&A.
Peer Assessment (Team Contribution)	ALL	10%	Evaluation by team members based on the separate rubric (Contribution, Reliability, Collaboration, scaled 1–10).

10. Peer Assessment

In this project, peer assessment will be conducted to evaluate individual contributions within the team.

Criteria	Low (1-4)	Medium (5-7)	High (8-10)
Contribution & Initiative	Avoided tasks. Contributed significantly less than others.	Reliably completed assigned tasks.	Proactive and self-motivated, producing high-quality work.
Reliability & Responsibility	Frequently missed deadlines;	Generally met deadlines and	Always met deadlines; work was

Criteria	Low (1-4)	Medium (5-7)	High (8-10)
	unprepared.	fulfilled requirements.	meticulous and trustworthy.
Collaboration & Communication	Did not share information; disrupted team harmony.	Participated in discussions and acted cooperatively.	Actively shared ideas, provided constructive feedback, fostered a positive environment.

Evaluation Method:

- Evaluate each team member anonymously (1–10 points).
- Individual Peer Score = (Average of all criterion scores received from all evaluators)
- Final Individual Score = Common Team Score (out of 90) + Individual Peer Score (out of 10)

11. Recommended Report Structure (Example Structure)

1. Cover Page (Title, Team Name, Member Names, IDs, AI Usage Declaration)
2. Abstract (Project objective, main design, conclusions)
3. Introduction (Background, problem statement, purpose)
4. Literature Survey & Improvement Concept
5. Sensor Design (Structure and Materials)
6. Prototyping Process (Printing conditions, problems, and solutions)
7. Evaluation Experiments & Data Analysis (Protocol, results, graphs)
8. Refinement Cycle (Comparison before and after improvements)
9. Comprehensive Discussion & Application Proposal
10. Conclusion
11. References (IEEE Style)
12. Appendices (Datasheets, raw data, CAD drawings, etc.)
    If you have any videos, you can upload it with other files.

Appendix: AI Usage Declaration Format (Example)

(Must be included on the report cover page)

In the preparation of this report, I have used AI tools as detailed below:

AI Tool Name	Purpose of Use	Section/Task Applied
(e.g., ChatGPT-4)	*(e.g., Grammar/style check)*	*(e.g., Abstract, Conclusion)*
(e.g., DeepL)	*(e.g., Translation to English)*	*(e.g., Entire text)*
(If none, please state "None")		

【Declarations for AI Usage】

- Any text generated directly by AI has been enclosed in quotation marks (" ") and cited with the AI tool name as the source.
- All experimental data and raw evaluation results were acquired solely by our team.
- The core components of the discussion, analysis, and conclusion were written in my own words without reliance on AI.
- I fully understand the contents of this report and take responsibility for explaining it orally if required.
- The proportion of AI-generated content does not exceed 10% of the entire work, in compliance with the module handbook.

Date: _____

Student Name (Signature/Print): _____

Student ID: _____